

Constraints on the contribution of local sources to the observed cosmic-ray electrons

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ABSTRACT

The recent observation of high-energy cosmic-ray electrons up to about 20 TeV by the H.E.S.S. experiment has revealed no significant features in the spectrum above around 1 TeV. This new observation has posed a severe challenge to the general view that nearby cosmic-ray sources can leave identifiable structures in the electron spectrum at multi-TeV energies. In this paper, we provide the maximum constraint imposed by the H.E.S.S. measurement on the contribution of the nearby supernova remnants to the observed cosmic-ray electrons in terms of the supernova kinetic energy injected into electrons and the escape time of cosmic rays from the remnants. We also show that the dipole anisotropy determined for the maximum contribution from the nearby remnants is well within the limits provided by the *Fermi*-LAT and the CALET experiments at GeV–TeV energies. Among the nearby remnants, we find Monogem and Vela emerging as the major contributors of cosmic rays above few TeVs. We explore the gamma-ray emission from Vela, and find the emission spectrum to be consistent with the constraint on the injection energy and the escape time of electrons obtained based on the observed cosmic-ray electron spectrum [We have to change this depending on what we get].

Key words. cosmic rays — diffusion — acceleration of particles — ISM: supernova remnants

1. Introduction

In the recent years, unprecedented high-precision measurements of the cosmic-ray (CR) electron spectrum up to ~ 20 TeV have become available, thanks to new-generation experiments such as the Alpha Magnetic Spectrometer (AMS-02; Aguilar et al. 2019a), Calorimetric Electron Telescope (CALET; Adriani et al. 2017), Dark Matter Particle Explorer (DAMPE; Ambrosi et al. 2017), Fermi Large Area Telescope (*Fermi*-LAT; Abdollahi et al. 2017a) and the High Energy Stereoscopic System (H.E.S.S.; Aharonian et al. 2008, 2009, 2024). These observations have revealed puzzling new features in the electron spectrum, including a subtle hardening around 40 GeV and a pronounced break or steepening near 1 TeV. Such features are difficult to reconcile with standard models of cosmic-ray acceleration and propagation in the Galaxy, which generally predict a simple power-law spectrum with an exponential cut-off. Moreover, the lack of significant spectral structures beyond ~ 1 TeV further complicates the issue as this is the energy region where contributions from nearby cosmic-ray sources are expected to leave distinct signatures in the observed spectrum (Kobayashi et al. 2004; Thoudam & Hörandel 2012).

For a given power-law source spectrum, the equilibrium spectrum of high-energy CR electrons in the Galaxy is mainly governed by radiative losses they suffer during their propagation through the Galaxy. Under the standard model of CR propagation where CR electrons undergo diffusive motion, and radiative losses are dominated by synchrotron and inverse Compton scattering processes, the equilibrium spectrum of electrons is expected to closely follow a smooth power-law of $E^{-\Gamma-1+a}$ for a continuous and stationary distribution of sources in the Galactic

disk (Ginzburg & Ptuskin 1976; Atoyan et al. 1995), where Γ is the source spectrum index and $a \sim (0.2 - 0.3)$ [Ankit: shouldn't the range be slightly wider, as we use 0.54(0.33) as the diffusion index (e.g., $a = (1 - 1/3)/2 = 1/3$ for $\delta = 1/3$)?] is small correction factor that depends on the index of the diffusion coefficient (see Section 2 for details). Although the exact origin of the observed spectral features is difficult to explain based on the standard model, the subtle hardening at ~ 40 GeV has been attributed to the change in the inverse Compton scattering cross-section of high-energy electrons from the Thomson limit to the Klein-Nishina regime on the ultraviolet background photons (Evoli et al. 2020), and the spectral break at ~ 1 TeV as due to a broken power-law electron source spectrum from supernova remnants resulting from the radiative losses suffered by high-energy electrons during their confinement in the downstream region of the shock before being released into the interstellar medium (Thoudam 2024). The presence of a broken power-law source spectrum for electrons has been predicted by simulation of diffusive shock acceleration by supernova shocks (Kang 2011), and also supported by the multi-wavelength studies of supernova remnants (Abdalla et al. 2018; Zang & Liu 2019).

Because of severe radiative losses, high-energy electrons cannot reach far distances in the Galaxy. The diffusion length of electrons¹ depends on the diffusion coefficient $D(E)$ and the energy loss time $t_{\text{loss}}(E)$ of the electrons as $r_d \propto \sqrt{D(E)t_{\text{loss}}(E)} \propto E^{(\delta-1)/2}$. For a typical value of $\delta = 0.33$ at high energies, it follows $r_d \propto E^{-0.33}$, which indicates a steep decrease in the diffusion length with energy. For the average magnetic field strength of $3 \mu\text{G}$ and the average energy densities of interstellar radiation fields, the diffusion length for electrons which finally arrive

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¹ The average distance traveled by a CR electron before losing its energy.

at the Earth with energy of 1 TeV is below ~ 1.7 kpc, as described in detail in Section 2. For electrons arriving with 5 TeV energy, r_d goes below ~ 1 kpc. This points towards the argument that the observed CR electrons in the TeV region are mostly produced by sources in the local region, while the lower energy region is dominated by the contribution from the distant sources. Under this consideration, the electron spectrum at low energies is expected to have a smooth behavior, while at high energies, especially above a few TeV, the spectrum is expected to show some distinct features (dips and bumps) due to the space-time discreteness of the very few local sources. If supernova remnants are the main sources of CRs in the Galaxy, as predicted by diffusive shock acceleration theory (Bell 1978; Drury 1983; Blandford & Eichler 1987) and supported by multi-wavelength observations of non-thermal emission from radio to very-high-energy gamma rays in several remnants (e.g., Bamba et al. 2006; Parizot et al. 2006; Abdalla et al. 2018 [These references need to be updated]), then the presence of nearby middle-aged remnants such as Monogem and Vela should naturally imprint identifiable features on the observed spectrum above a few TeV. However, the absence of any significant spectral features in the observed spectrum above ~ 1 TeV pose a severe challenge to this general view.

The observed smooth behavior of the spectrum imposes a strong constraint on the contribution of nearby supernova remnants to the observed CR electrons in the TeV region. In this work, we focus on obtaining the constraint in terms of the amount of supernova kinetic energy injected into CR electrons and the time-dependent nature of CR escape from the remnant. We consider that the bulk of the observed CRs up to a few TeV are produced by distant sources (hereafter, "background CRs") and the nearby sources can have appreciable contribution only close to the highest energies observed (hereafter, "nearby CRs"). We also determine the CR electron dipole anisotropy corresponding to the maximum contribution of the nearby supernova remnants (SNRs), and compare it with the limits from the *Fermi*-LAT and the CALET experiments above ~ 50 GeV. The constraint on the electron injection energy is compared with the particle energies required to explain the non-thermal emissions from the Vela and Monogem supernova remnants.

The paper is planned as follows. In Section 2, the diffusion equation governing the propagation of CR electrons is described, separately for the background and the nearby CRs. In Section 3, the fit to the observed spectrum with the background CRs is presented and the maximum contribution of the nearby supernova remnants is obtained. In Section 5, the main results of the present work is presented along with a discussion on the dipole anisotropy and non-thermal emissions from few of the nearby supernova remnants. Finally, in Section 6, we present the main conclusion from this work.

2. Propagation of cosmic-ray electrons

Cosmic rays undergo diffusive propagation in the Galaxy due to scattering by magnetic field irregularities or by self-excited Alfvén and hydromagnetic waves in the interstellar medium (ISM), which are generated predominantly by the nuclei component of CRs [Ref. ???]. For high-energy electrons, the propagation is also associated with radiative losses mainly due to synchrotron emission in the presence of the Galactic magnetic field and inverse Compton processes in the ambient photon fields.

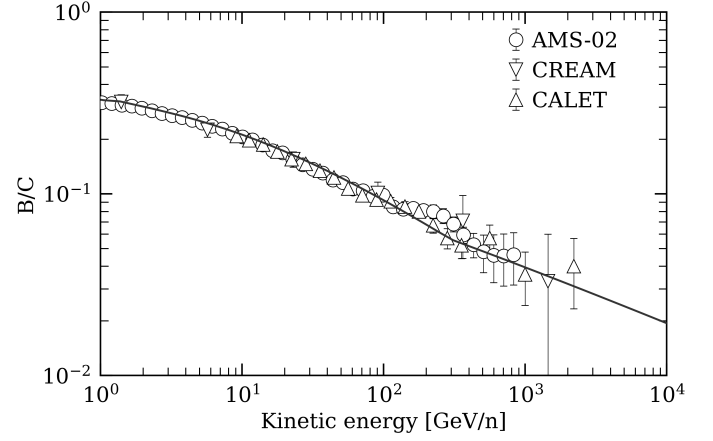


Fig. 1: Boron-to-carbon ratio (AMS-02, CREAM, CALET) compared with the model using a break in $D(E)$ at $E_1 = 300$ GeV/n.

2.1. Cosmic-ray diffusion coefficient

The diffusion coefficient is usually obtained using the observed secondary-to-primary ratios, in particular, boron-to-carbon ratio (B/C). Recent measurements of B/C by the AMS-02, CALET and DAMPE experiments seem to indicate a hardening of the ratio above a few hundred GeV/nucleon (Aguilar 2016; Adriani 2022; DAMPE Collaboration 2022). Under a pure diffusion model, such a feature can be explained by introducing a break in the diffusion coefficient of CRs (Génolini et al. 2017). However, in more sophisticated models of CR propagation that include CR reacceleration by the turbulent magnetic field or by weak shocks in the Galaxy, the observed hardening in the B/C ratios is naturally explained without invoking a break in the diffusion coefficient (e.g., Seo & Ptuskin 1994; Thoudam & Hörandel 2014). For CR electrons, the effect of reacceleration on the spectrum is subdominant compared to that of energy losses during propagation. Therefore, in the present study which focuses only on CR electrons, we adopt a pure diffusion model, with the diffusion coefficient assumed to have the following form (Bhadra et al. 2025),

$$D(\rho) = D_0 \beta \times \begin{cases} \left(\frac{\rho}{\rho_0}\right)^{\delta_1}, & \rho \leq \rho_1, \\ \left(\frac{\rho_1}{\rho_0}\right)^{\delta_1} \left(\frac{\rho}{\rho_1}\right)^{\delta_2}, & \rho > \rho_1, \end{cases} \quad (1)$$

where ρ denotes the particle rigidity, D_0 is the diffusion constant, β is the particle velocity in units of c , $\rho_0 = 3$ GV is a constant and $\delta_1(\delta_2)$ is the diffusion index below (above) the break rigidity ρ_1 . The rigidity is defined as $\rho = Apc/Ze$, where A and Z are the mass and charge numbers of the CR particle respectively, e is the charge of an electron, c the velocity of light and p represents the momentum per nucleon of the particle. By optimizing B/C calculated using Equation 1 to the observed B/C data, we obtain $D_0 = 1.55 \times 10^{28} \text{ cm}^2 \text{ s}^{-1}$, the break energy (in energy per nucleon) corresponding to the break rigidity ρ_1 as $E_1 = 300$ GeV/n, $\delta_1 = 0.54$ and $\delta_2 = 0.33$. Our calculation of B/C follows the procedure described in Thoudam & Hörandel 2013 which considers inelastic collisions of CR nuclei with the interstellar matter in addition to the diffusive motion, a uniform distribution of CR sources in the Galactic disk and a Galactic halo height of 5 kpc. Figure 1 compares our B/C result with the

measurements from the AMS-02, CALET and CREAM experiments.

2.2. Radiative energy-loss rate of electrons

For the present work which focuses on CR electrons above ~ 10 GeV, the total energy loss of electrons is considered to be due to synchrotron radiation and inverse-Compton scattering processes as $b(E) = b_{\text{syn}}(E) + b_{\text{IC}}(E)$. The synchrotron loss rate as a function of energy E is given by,

$$b_{\text{syn}}(E) = \frac{4}{3} c \sigma_T U_B \left(\frac{E}{m_e c^2} \right)^2, \quad (2)$$

where σ_T is the Thomson cross-section, $U_B = B^2/8\pi$ is the magnetic-field energy density, and $m_e c^2$ is the electron rest mass energy. We adopt the average magnetic field strength evaluated over a cylindrical region centered at the Sun with radius R and half-height H , using the magnetic field distribution $B(r, z) = B_0 e^{-(r-R_0)/R_B} e^{-|z|/z_B}$, where $R_0 = 8$ kpc denotes the distance of the Sun from the Galactic center, $R_B = 50$ kpc, and $z_B = 3$ kpc, as implemented in the GALPROP CR propagation code (Strong et al. 2010). We take $B_0 = 6 \mu\text{G}$, corresponding to the local magnetic field strength reported by Beck 2001. For $R = 5$ kpc and $H = 5$ kpc (R5H5), we obtain an average magnetic field $B = 2.90 \mu\text{G}$. For the smaller region with $R = 2$ kpc and $H = 2$ kpc (R2H2), the average field increases to $B = 4.37 \mu\text{G}$, as expected from the steep decline of the field with z .

An accurate description of the interstellar radiation field (ISRF) is essential for evaluating inverse-Compton (IC) energy losses. We adopt the three-dimensional radiative-transfer model of Popescu et al. (2017), which provides a self-consistent representation of the Milky Way's ultraviolet-to-submillimetre radiation field based on *COBE*, *IRAS*, and *Planck* all-sky data. The model uses an axisymmetric geometry to describe the radial and vertical distributions of stellar emissivity and dust opacity and tabulates the resulting stellar and dust radiation fields on a cylindrical (r, z) grid. The ISRF data are obtained from the CDS VizieR catalogue². To characterize the photon field in the local Galactic environment, we compute the average ISRF within the same cylindrical regions (R2H2 and R5H5). The resulting mean ISRF spectra are fitted with a multi-gray-body model to obtain analytic representations of the stellar and dust components. The derived gray-body parameters (T_i , w_i) for both averaging regions are listed in Table 1 and the cosmic microwave background (CMB) component is adopted from Fixsen (2009).

For computational efficiency, we calculate the inverse-Compton loss rate using the parameterization provided in Fang et al. (2021) instead of the exact Klein-Nishina cross-section (Jones 1965; Strong & Moskalenko 1998) as,

$$b_{\text{IC}}(E) = \frac{20 c \sigma_T}{\pi^4} \left(\frac{E}{m_e c^2} \right)^2 \sum_i w_i Y(E, T_i), \quad (3)$$

where the summation is over the different photon fields listed in Table 1 and $Y(E, T)$ is a function that depends on the energy of the incident electron and the effective temperature of the photon field. Writing $x = 4EkT/(m_e c^2)^2$, the function $Y(E, T) \equiv Y(x)$ is

given by,

$$Y(x) = \begin{cases} \frac{\pi^4}{15}, & x \leq 1.5 \times 10^{-3}, \\ \exp \left[\sum_{j=0}^6 c_j (\ln x)^j \right], & 1.5 \times 10^{-3} < x < 150, \\ \frac{3}{4} \left(\frac{\pi}{x} \right)^2 (\ln x - 1.9805), & x \geq 150, \end{cases} \quad (4)$$

with the coefficients c_j provided in Appendix A. Equation 3 reproduces the loss rate given by the full Klein-Nishina cross-section to within 1% accuracy over the energy range of our interest.

The total IC energy-loss rate for relativistic electrons, computed using Equation 3, is shown in Figure 2 for the two ISRF averaging volumes (R2H2 and R5H5). The differences between these cases illustrate the sensitivity of IC losses to the local radiation energy density: smaller averaging regions with higher ISRF intensities produce slightly larger loss rates, while more extended regions with lower average ISRF intensities result in weaker IC losses. For comparison, Figure 2 also includes the loss rates adopted by Fang et al. (2021) and Schlickeiser & Ruppel (2010). The IC loss rates computed in this work are lower than those shown in Fang et al. (2021). For a direct comparison, we highlight the R2H2 case, which employs the same averaging region as their calculation. The lower IC loss rate arises because the ISRF of Popescu et al. (2017) adopted here results in a lower local radiation energy density than the ISRF of Porter et al. (2008) used by Fang et al. (2021). As discussed by Popescu et al. (2017), their radiative-transfer model has been shown to better match UV-MIR observations of the Milky Way, largely due to its more detailed treatment of the stellar-dust geometry and locally heated dust emission. The corresponding synchrotron losses are also shown for magnetic field strengths averaged over the same regions ($B = 4.37 \mu\text{G}$ for R2H2 and $B = 2.90 \mu\text{G}$ for R5H5). At high energies, the synchrotron losses become comparable to or exceed the IC component, depending on the magnetic field strength, whereas IC losses dominate at lower energies.

2.3. Diffusion length of high-energy electrons

Compared to nuclear CR species, high-energy electrons travel only limited distances in the Galaxy due to radiative losses. This is expected to be severe mostly for electrons in the multi-TeV regime. [Compared to nuclear cosmic ray species, high-energy electrons travel much shorter distances in the Galaxy due to strong radiative losses at multi-TeV energies-Ankit:can we write this instead?]. In the presence of radiative losses, the diffusion length of high-energy electrons with energy E' before it cools down to energy E can be calculated using $r_d(E, E') = 2 \left[\int_E^{E'} \{D(u)/b(u)\} du \right]^{1/2}$, where the diffusion coefficient D is given by Equation 1, and the total energy loss rate b is the sum of the synchrotron and inverse-Compton loss rates as given by Equations 2 and 3 respectively. Figure 3 shows the diffusion lengths of electrons with initial energies up to $E' = 100$ TeV before they cool to final energies of $E = (10, 100)$ GeV and $(1, 5)$ TeV. CR electrons observed at the Earth at a given energy include contributions from electrons injected with different initial energies by sources located at different distances. Both diffusive shock acceleration theory applied to supernova shocks in a typical interstellar medium as well as multi-wavelength studies of young supernova remnants in the Galaxy suggest that electrons can be accelerated up to ~ 100 TeV (Reynolds & Keohane

² <https://cdsarc.cds.unistra.fr/viz-bin/cat/J/MNRAS/470/2539>

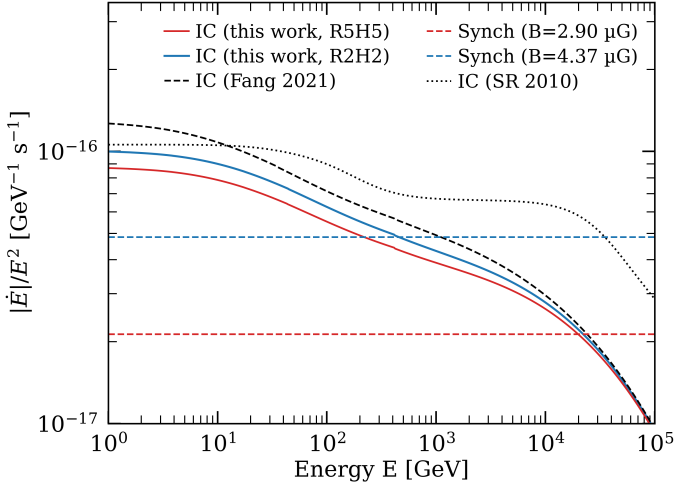


Fig. 2: Inverse-Compton (IC) and synchrotron energy-loss rates for relativistic electrons, shown as $(\dot{E}/E^2)$ versus energy. The solid lines represent the IC losses computed in this work for the ISRF models R2H2 (blue), and R5H5 (red). For comparison, the black dashed line shows the IC loss rate from Fang et al. (2021), while the dotted line represents the result from Schlickeiser & Ruppel (2010). The horizontal dashed lines indicate synchrotron losses for magnetic fields of $B = 4.37 \mu\text{G}$ and $B = 2.90 \mu\text{G}$, corresponding to averages over 2 kpc and 5 kpc regions, respectively.

Table 1: Parameters obtained from our grey-body fits to the local mean ISRF. The stellar and dust emission data used in the fit are taken from the radiative-transfer analysis of Popescu et al. (2017) and averaged within cylinders of radius and half-height 5 kpc (R5H5) and 2 kpc (R2H2) centred on the Earth. We adopt R5H5 as our default ISRF model. Listed are the temperatures T_i (K) and energy densities W_i (eV cm^{-3}) of the stellar, dust, and CMB components.

i	R5H5		R2H2	
	T_i [K]	W_i [eV cm^{-3}]	T_i [K]	W_i [eV cm^{-3}]
<i>Stellar components</i>				
1	20521.83	1.80×10^{-2}	20650.37	2.37×10^{-2}
2	4815.69	1.74×10^{-1}	4915.28	2.04×10^{-1}
3	2580.28	2.43×10^{-1}	2520.79	2.83×10^{-1}
4	708.52	1.73×10^{-2}	710.31	2.25×10^{-2}
<i>Dust components</i>				
5	292.73	2.51×10^{-2}	294.08	3.44×10^{-2}
6	64.80	2.27×10^{-2}	64.71	2.86×10^{-2}
7	26.25	1.09×10^{-1}	25.28	1.44×10^{-1}
<i>CMB</i>				
8	2.725	2.60×10^{-1}	2.725	2.60×10^{-1}

1999; Reynolds 2008). From Figure 3, we can see that electrons detected at 10 GeV energy can originate from distances as large as ~ 5 kpc (for initial electron energies of 100 TeV), whereas electrons detected at TeV energies predominantly arise from sources much closer to the Earth. For instance, electrons observed at 1 TeV and 5 TeV are mostly produced within ~ 1.5 kpc

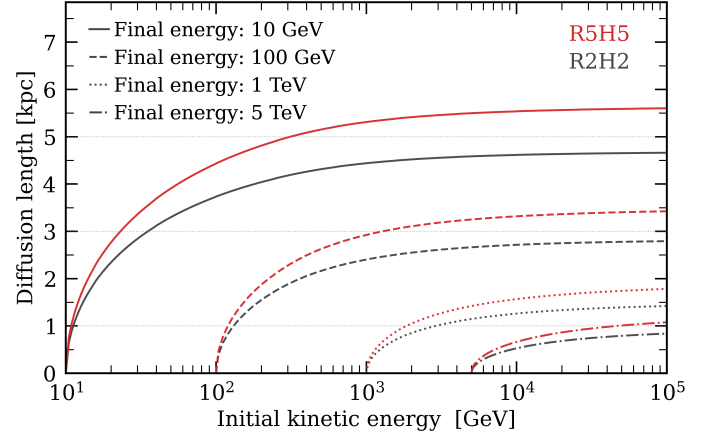


Fig. 3: Diffusion length $r_d(E, E')$ as a function of initial kinetic energy for different final energies, as described in Section 2.3. The calculation accounts for energy losses due to inverse Compton scattering due to interstellar radiation field (ISRF) and synchrotron emission in the Galactic magnetic field, both averaged over cylindrical regions centered on the Sun with radius $R = 2$ kpc and half-height $H = 2$ kpc (R2H2, black) and $R = 5$ kpc and $H = 5$ kpc (R5H5, red). Solid, dashed, dotted, and dash-dotted curves correspond to final energies of 10 GeV, 100 GeV, 1 TeV, and 5 TeV, respectively.

and ~ 1 kpc respectively. This implies that above ~ 5 TeV, the observed electrons becomes increasingly sensitive to the discrete space-time distribution of nearby CR sources.

In the following, motivated by the study of Thoudam 2024 which demonstrated that the observed break in the all-electron spectrum at ~ 1 TeV is best explained by a broken power-law source spectrum of electrons from distant sources (background CRs), and by the absence of any distinct spectral feature above a few TeV, we model the observed CR electrons as being dominated by background CRs up to ~ 5 TeV, while at higher energies, we consider the contribution from nearby sources (nearby CRs) to become increasingly significant.

2.4. Propagation of background cosmic-ray electrons

The background component of CR electrons is modeled as the diffuse contribution from distant supernova remnants (SNRs) that continuously inject particles into the interstellar medium. Their transport is described by the steady-state diffusion-loss equation,

$$\nabla \cdot (D \nabla N) + \frac{\partial}{\partial E} (b(E) N) = -Q, \quad (5)$$

where $N(E, \mathbf{r})$ is the differential number density of electrons at energy E and position $\mathbf{r} \equiv (r, \phi, z)$ in cylindrical coordinates, with r the distance from the Sun, ϕ the azimuthal angle, and z the height above the Galactic plane. $D(E)$ is the diffusion coefficient, and $b(E) = -dE/dt$ is the total energy-loss rate from synchrotron radiation and inverse-Compton scattering. The source term $Q(E, \mathbf{r})$ describes the steady injection rate of electrons per unit energy and volume into the ISM.

As high-energy electrons cannot travel large distances in the Galaxy due to radiative losses, the influence of the Galactic halo boundary on the solution of Equation (5) can be neglected within our region of interest for energies above ~ 10 GeV. Under this

approximation, the Green's function solution to Equation (5) is obtained as

$$G(E, E', \mathbf{r}, \mathbf{r}') = \frac{1}{8\pi^{3/2}b(E)A^{3/2}} \exp\left[-\frac{(\mathbf{r} - \mathbf{r}')^2}{4A}\right], \quad (6)$$

where

$$A(E, E') = \int_E^{E'} \frac{D(u)}{b(u)} du, \quad (7)$$

The CR electron density at position $\mathbf{r}$ is then calculated as

$$N(\mathbf{r}, E) = \int_E^\infty dE' \int d^3r' G(\mathbf{r}, \mathbf{r}', E, E') Q(\mathbf{r}', E'), \quad (8)$$

where, for a uniform source distribution in the Galactic disk and assuming azimuthal symmetry, the source term is written as $Q(\mathbf{r}, E) = \nu q(E) \delta(z)$. Here, ν is the supernova explosion rate per unit surface area in the disk and $q(E)$ is the source spectrum. The factor $\delta(z)$ confines the sources to the Galactic plane, consistent with the thin-disk approximation commonly used in CR transport models. A value of $\nu = 25 \text{ Myr}^{-1} \text{ kpc}^{-2}$ is adopted, corresponding to about three supernova explosions per century in the Galaxy (Grenier 2000). Evaluating Equation (8) at $\mathbf{r} = 0$, the CR electron density at Earth due to sources beyond a radial distance r_0 in the Galactic disk can be written as

$$\begin{aligned} N(E) &= 2\pi\nu \int_E^\infty dE' \int_{r_0}^\infty r' dr' \int_{-\infty}^\infty dz' G(r', z'; E, E') q(E') \delta(z') \\ &= \frac{\nu}{\sqrt{\pi} b(E)} \int_E^\infty \frac{q(E')}{r_d} \exp\left(-\frac{r_0^2}{r_d^2}\right) dE', \end{aligned} \quad (9)$$

where $r_d(E, E') = 2\sqrt{A(E, E')}$ represents the average diffusion length that an electron injected with energy E' can travel before cooling to energy E .

2.5. Propagation of cosmic-ray electrons from nearby sources

A spatially uniform source distribution provides a reasonable approximation for distant sources contributing to the background cosmic rays (CRs). In contrast, nearby sources must be treated as discrete objects, since their finite size, morphology, and temporal evolution can strongly affect the CR electron flux observed at the Earth (Thoudam & Hörandel 2012). Their contribution is described by the time-dependent diffusion–loss equation

$$\frac{\partial N}{\partial t} = \nabla \cdot [D(E) \nabla N] + \frac{\partial}{\partial E} [b(E) N] + Q(E, \mathbf{r}, t), \quad (10)$$

where $N(E, \mathbf{r}, t)$ is the differential number density of electrons at position $\mathbf{r} \equiv (r, \phi, z)$ and time t . We adopt cylindrical coordinates (r, ϕ, z) centred on the Solar position in the Galactic plane, where r is the radial distance from the Sun in the plane, ϕ is the azimuthal angle, and z is the height above the Galactic plane. The quantity E denotes the kinetic energy, $D(E)$ is the diffusion coefficient, and $b(E) = -dE/dt$ is the total radiative energy-loss rate, accounting for synchrotron losses in the Galactic magnetic field and inverse-Compton scattering off the interstellar radiation field. The source term $Q(E, \mathbf{r}, t) = q(E) q(\mathbf{r}) q(t)$ describes the injection rate of electrons per unit volume into the interstellar medium (ISM). The Green's function of Eq. (10) can be written as (see, e.g. Thoudam & Hörandel 2012)

$$G(E, E', \mathbf{r}, \mathbf{r}', t, t') = \frac{1}{8(\pi A)^{3/2} b(E)} \exp\left[-\frac{(\mathbf{r} - \mathbf{r}')^2}{4A}\right] \times \delta(t' - t + \tau(E, E')), \quad (11)$$

where,

$$A(E, E') = \int_E^{E'} \frac{D(u)}{b(u)} du, \quad \tau(E, E') = \int_E^{E'} \frac{1}{b(u)} du. \quad (12)$$

The CR electron density at position $\mathbf{r}$ and time t is obtained by convolving the Green's function with the source term as,

$$N(\mathbf{r}, E, t) = \int_E^\infty dE' \int d^3r' \int_{-\infty}^t dt' G(E, E', \mathbf{r}, \mathbf{r}', t, t') \times Q(E', \mathbf{r}', t'), \quad (13)$$

The escape of cosmic rays from SNRs, which determines the form of $Q(E, \mathbf{r}, t)$, is governed by the interplay between diffusive confinement and the dynamical evolution of the shock. As the shock expands and decelerates, particles with sufficiently large diffusion lengths can no longer remain confined near the shock front and progressively escape into the surrounding medium. The maximum energy of the escaping particles is therefore governed by the temporal evolution of the magnetic field and the shock velocity, which together determine the efficiency of particle confinement and acceleration. These quantities evolve through complex and poorly constrained processes such as magnetic-field amplification, turbulence generation, and feedback from the accelerated CR population. Because of these uncertainties, the time-dependent escape of CRs is modeled by parameterizing the escape energy as (e.g. Gabici et al. 2009; Ohira et al. 2011; Thoudam & Hörandel 2012)

$$E_{\text{esc}}(t) = E_{\text{max}} \left(\frac{t}{t_{\text{sed}}} \right)^{-\alpha}, \quad (14)$$

where E_{max} denotes the energy of the highest-energy CRs released at the onset of the Sedov phase t_{sed} , and α is a parameter that describes the nature of CR escape from the remnant. We adopt $E_{\text{max}} = 1 \text{ PeV}$ which corresponds to the maximum energy of CR protons predicted by diffusive shock acceleration theory applied to SNRs (Bell 2004) ([Put references here. You may find them in my old papers]), and a Sedov time of $t_{\text{sed}} = 1000 \text{ yr}$ for all nearby SNRs (see Appendix B). The corresponding escape time for particles of energy E is obtained from Equation (14) as

$$t_{\text{esc}}(E) = t_{\text{sed}} \left(\frac{E}{E_{\text{max}}} \right)^{-1/\alpha}. \quad (15)$$

At a later stage of the SNR evolution when the shock can no longer efficiently accelerate particles, the turbulence level in the upstream cannot effectively confine the particles. At this stage, all remaining CRs are assumed to be released into the interstellar medium (ISM). To represent this transition, we impose an upper limit of 10^5 yr on the escape time, defining $T_{\text{esc}}(E) = \min[t_{\text{esc}}(E), 10^5 \text{ yr}]$ (Berezhko & Völk 2000). The radius of the SNR at the epoch when CRs of energy E escape into the ISM, as given by Sturmer et al. (1997), is

$$R_{\text{esc}}(E) = 2.5 v_0 t_{\text{sed}} \left[\left(\frac{T_{\text{esc}}}{t_{\text{sed}}} \right)^{0.4} - 0.6 \right], \quad (16)$$

where v_0 denotes the initial shock velocity, taken to be $v_0 = 3.5 \times 10^3 \text{ km s}^{-1}$ as adopted in our model (see Appendix B).

Assuming a spherically symmetric remnant and a uniform distribution of CRs over its surface prior to release, the source term is then expressed as

$$Q(r, E, t) = \frac{q(E)}{4\pi R_{\text{esc}}^2} \delta(r - R_{\text{esc}}) \delta(t - T_{\text{esc}}), \quad (17)$$

where $q(E)$ represents the injection spectrum of CRs at energy E . For a SNR located at a distance r_s from Earth, the differential number density of electrons arriving at Earth with energy E can be expressed in terms of their injection energy E_i , which is the energy at which they escaped the SNR before cooling during their propagation to Earth. Substituting the source term into Equation (13) gives

$$N(r_s, E, t_s) = \frac{q(E_i) b(E_i)}{8 [\pi A(E, E_i)]^{3/2} b(E)} \exp\left[-\frac{r_s^2 + R_{\text{esc}}(E_i)^2}{4 A(E, E_i)}\right] \times \frac{\sinh(\xi)}{\xi}, \quad (18)$$

where $\xi = r_s R_{\text{esc}}(E_i) / [2 A(E, E_i)]$. In this expression, $A(E, E_i)$ is defined in Equation (12), $q(E_i)$ represents the source spectrum at the injection energy, and $R_{\text{esc}}(E_i)$ denotes the SNR radius at the time electrons of energy E_i were released. The factor $\sinh(\xi)/\xi$ accounts for the geometrical correction arising from the finite size of the spherical shell. The injection energy E_i is related to the observed energy E through the cooling condition $t_s - t_{\text{esc}}(E_i) = \tau(E, E_i)$, where t_s is the SNR age and $t_{\text{esc}}(E_i)$ the corresponding escape time. In the limit $R_{\text{esc}} \rightarrow 0$, Equation (18) reduces to the well-known point-source solution for a burst-like injection (Thoudam & Hörandel 2012).

3. Fitting procedure of the observed all-electron spectrum

With the framework of the transport equation presented in Section 2, we determine the CR electron source parameters by fitting our model predictions to the all-electron ($e^- + e^+$) flux measured by the DAMPE and the H.E.S.S. experiments. We begin by focusing on the electron background component produced by distant SNRs. This background is expected to dominate the observed spectrum up to a few TeV, before the influence of nearby sources becomes significant. In this work, the fit to the background component is restricted to the energy range of 20 GeV–5 TeV. In the next step, we assess the maximum contribution allowed from nearby SNRs using the measured data at higher energies available up to ~ 40 TeV.

3.1. Fit with the background cosmic-ray electrons

Motivated by the analysis of Thoudam (2024), who showed that the ~ 1 TeV break in the cosmic-ray electron spectrum is best reproduced with a broken power-law injection spectrum, we compute the background electron flux as described in Section 2.4. For the source term $q(E)$, we adopt a smooth broken power-law of the form,

$$q(E) = q_0 E^{\Gamma_1} \left[1 + \left(\frac{E}{E_b} \right)^{1/\delta} \right]^{\delta(\Gamma_2 - \Gamma_1)}, \quad (19)$$

where q_0 is the normalization of the injected electron spectrum, Γ_1 and Γ_2 are the spectral indices below and above the break

energy E_b , and the smoothness parameter is fixed to $\delta = 0.05$ (Thoudam 2024). The normalization q_0 is related to the electron injection efficiency, which is defined as the fraction f_{bg} of the supernova kinetic energy $E_{\text{SN}} = 10^{51}$ erg transferred to electrons above 1 GeV. We consider two representative forms for the source spectrum. In the first case (hereafter Model I), the relation $\Gamma_2 = \Gamma_1 - 1$ is imposed, reflecting the steepening expected for electrons continuously injected and subject to radiative cooling downstream of the shock (Kang 2011; Thoudam 2024). In the second case (hereafter Model II), Γ_2 is left free in order to evaluate how the choice of post-break slope affects the background fit and the resulting constraints on nearby-source contributions. For both Model I and Model II, the source parameters, together with the radial distance r_0 introduced in Equation 9, are treated as free parameters. The corresponding parameter sets are

$$\theta_{\text{I}} = (f_{\text{bg}}, \Gamma_1, E_b, r_0), \quad \theta_{\text{II}} = (f_{\text{bg}}, \Gamma_1, \Gamma_2, E_b, r_0). \quad (20)$$

We model the all-electron spectrum $I_{\text{M}}(E)$ as the sum of the contributions of the electron and positron fluxes, $I_{\text{M}}(E) = I_{e^-}(E) + I_{e^+}(E)$. The electron flux is calculated as $I_{e^-}(E) = (\nu/4\pi) N(E)$ where $N(E)$ is given by Equation 9 and $\nu = c\beta(E)$ denotes the particle velocity, and the positron flux is adopted from the empirical parametrization provided by the AMS-02 collaboration (Aguilar et al. 2019b),

$$I_{e^+}(E) = \frac{E^2}{\hat{E}^2} \left[K_1 \left(\frac{\hat{E}}{E_1} \right)^{\gamma_1} + K_2 \left(\frac{\hat{E}}{E_2} \right)^{\gamma_2} \exp\left(-\frac{\hat{E}}{E_c}\right) \right], \quad (21)$$

where $\hat{E} = E + \phi$ accounts for solar modulation, with $\phi = 1.1$ GeV being the modulation potential. The parameter values used are $K_1 = 6.51 \times 10^{-2} (\text{m}^2 \text{ sr s GeV})^{-1}$, $K_2 = 6.80 \times 10^{-5} (\text{m}^2 \text{ sr s GeV})^{-1}$, $E_1 = 7.0$ GeV, $E_2 = 60$ GeV, $\gamma_1 = -4.07$, $\gamma_2 = -2.58$, and $E_c \approx 813$ GeV. The first term of the equation 21 represents the diffuse positron component produced by secondary interactions of cosmic-ray nuclei with the interstellar medium, whereas the second term accounts for an additional positron contribution that may originate from dark-matter annihilation or from astrophysical sources such as Galactic pulsars.

To constrain the model parameters, we perform Bayesian inference using a Markov Chain Monte Carlo (MCMC) approach. The likelihood is defined through the χ^2 statistic,

$$\chi^2 = \sum_{j,i} \frac{[I_{\text{obs}}^{(j)}(E_i) - I_{\text{M}}(E_i)]^2}{\sigma_{(j)}^2(E_i)}, \quad (22)$$

where $I_{\text{obs}}^{(j)}$ and $\sigma_{(j)}$ denote the measured flux and associated uncertainty in the i -th energy bin of experiment j , and the likelihood is proportional to $\exp(-\chi^2/2)$. Posterior sampling is carried out with the affine-invariant sampler *emcee* (Foreman-Mackey et al. 2013) using 30 walkers and a hybrid proposal strategy combining *DEMove* (80%) and *DESnookeyMove* (20%) (Foreman-Mackey et al. 2019). Uniform (flat) priors are considered for all parameters, and convergence of the posterior samples is assessed using the Gelman–Rubin statistic ($\hat{R}$).

³ The Gelman–Rubin convergence diagnostic ($\hat{R}$) assesses whether multiple MCMC chains are sampling a common posterior distribution. Values close to unity indicate good mixing and reliable convergence (Vehtari et al. 2021; Kumar et al. 2019). We compute $\hat{R}$ using the implementation provided in the *ArviZ* Python library.

Table 2: Priors, taken to be uniform within the ranges shown, posterior parameter estimates given as the median with $\pm 1\sigma$ highest-density intervals (HDIs), and best-fit values for all parameters in Model I and Model II. Results are from fits to the background cosmic-ray electron spectra in the averaging regions R5H5 and R2H2. The χ^2 and reduced χ^2_ν values refer to the best-fit solutions: for Model I (four free parameters, degree of freedom = 41) we obtain $\chi^2 = 77.10$ ($\chi^2_\nu = 1.88$) in R2H2 and $\chi^2 = 77.30$ ($\chi^2_\nu = 1.89$) in R5H5; for Model II (five free parameters, degree of freedom = 40) both regions yield $\chi^2 = 44.21$ ($\chi^2_\nu = 1.11$).

Parameter	Prior	Model I				Model II			
		R5H5		R2H2		R5H5		R2H2	
		Posterior	Best fit	Posterior	Best fit	Posterior	Best fit	Posterior	Best fit
$f_{\text{bg}} [\times 10^{48} \text{ erg}]$	[0, 10]	$1.48^{+0.04}_{-0.03}$	1.46	$1.71^{+0.04}_{-0.04}$	1.70	$1.46^{+0.03}_{-0.03}$	1.47	$1.70^{+0.04}_{-0.04}$	1.70
Γ_1	[-3, -2.0]	$-2.39^{+0.02}_{-0.02}$	-2.39	$-2.37^{+0.02}_{-0.02}$	-2.38	$-2.43^{+0.01}_{-0.01}$	-2.43	$-2.42^{+0.01}_{-0.01}$	-2.42
Γ_2	[-5, -2]	—	—	—	—	$-3.82^{+0.08}_{-0.08}$	-3.84	$-3.81^{+0.08}_{-0.08}$	-3.81
$E_b [\text{GeV}]$	[500, 20000]	1682^{+160}_{-120}	1625	1688^{+159}_{-124}	1654	1651^{+91}_{-76}	1615	1648^{+90}_{-73}	1601
$r_0 [\text{pc}]$	[0, 3000]	256^{+65}_{-54}	232	208^{+52}_{-45}	194	$47.32^{+51.97}_{-33.53}$	3.55	$38.19^{+41.35}_{-26.80}$	1.88

3.2. Adding the maximum contribution from nearby sources

For nearby SNRs, we adopt the same functional form for the source spectrum as for the background component, given by Equation (19). The spectral indices (Γ_1, Γ_2) and the break energy E_b are fixed to the best-fit values obtained from the background fit for the corresponding model (Model I or Model II). The source normalization, which is related to the nearby electron injection efficiency f_{near} , and the particle escape-time parameter α are treated as free parameters. All nearby SNRs are assumed to share common values of f_{near} and α .

We consider only those nearby SNRs located within a radial distance R_{near} from the Earth which is inferred from the fit parameter r_0 obtained from the background fit (see Section 4.1). The electron flux from a single nearby remnant j is computed as $I_{\text{e}}^j(r_j, E, t_j) = (v/4\pi) N(r_j, E, t_j)$, where r_j and t_j denote the distance and age of the remnant, $N(r_j, E, t_j)$ is given by Equation (18), and v is the particle velocity. Then, the total nearby electron flux is obtained by summing over all the nearby remnants as,

$$I_{\text{near}}(E) = \sum_j I_{\text{e}}^j(r_j, E, t_j). \quad (23)$$

To determine the maximum contribution of the nearby SNRs allowed by DAMPE and H.E.S.S. measurements, we construct an empirical upper bound of the measured flux based on the 3σ statistical uncertainties of the data points that lie above the background flux predicted by our model. The upper bound is extended to the highest measured energies by including the upper limits on the flux reported by the H.E.S.S. measurements up to ~ 40 TeV. A continuous upper bound curve in energy is generated by interpolating the upper-bound points. After the background electron flux is fixed, as described in Section 4.1, the nearby flux is added to it, and the total background plus nearby flux predicted by our model is compared with the 3σ upper bound to determine the maximum contribution of nearby SNRs allowed by current measurements.

Because the exact nature of particle escape from SNRs is poorly known, we treat the escape-time parameter α (see Equation 15) as a free parameter and explore a broad range of its possible values. We consider 10^5 realizations of α values uniformly distributed between 0.2 and 3.0. The lower value of $\alpha = 0.2$ corresponds to the case where all the CR particles (except for

the highest energy ones) remain confined until the end of the SNR evolution which is taken to be 10^5 years, and $\alpha = 3.0$ represents the other extreme where all the particles with energies above ~ 10 GeV escape the remnant at early stages close to the Sedov time which is considered to be 10^3 years (see Appendix B). For each value of α , we determine the maximum value of CR electron injection efficiency f_{near} for which the sum of the background and nearby electron fluxes does not exceed the constructed 3σ upper bound of the measured flux at any given energy. Different α values correspond to different particle escape times which leads to different total nearby fluxes at the Earth as shown in Appendix C.1. This leads to different corresponding maximum values of f_{near} for different values of α .

4. Results of the fit

We present below the results of the fit on the background CR electrons and the maximum contribution of the nearby SNRs to the observed all-electron spectrum.

4.1. Background cosmic-ray electrons

The background spectrum is fitted to the measured all-electron spectrum from DAMPE and H.E.S.S. experiments in the range of 20 GeV–5 TeV. The fits are performed for Model I and Model II, as described in Section 3.1, considering the electron energy loss rates obtained for the two different Galactic averaging regions (R5H5 and R2H2) introduced in Section 2.2. The fit results for the model parameters ($f_{\text{bg}}, \Gamma_1, \Gamma_2, E_b, r_0$) are given in Table 2, which lists for each parameter the prior range, the posterior median with uncertainties corresponding to the 1σ credible interval, and the maximum-a-posteriori (MAP) estimate as the best-fit value⁴. All the fit parameters satisfy $\hat{R} < 1.01$, indicating excellent convergence. The fit results for the R5H5 and R2H2 cases are shown in Figures 4 and 5 respectively. In each figure, the top panels correspond to Model I, while the bottom panels correspond to Model II. The left panels show the resulting background spectrum from the fit along with the fit residuals,

⁴ In Bayesian framework, the MAP estimate corresponds to the parameter values that maximize the posterior probability distribution and, for flat priors, coincides with the maximum-likelihood estimate which represents the best-fit value.

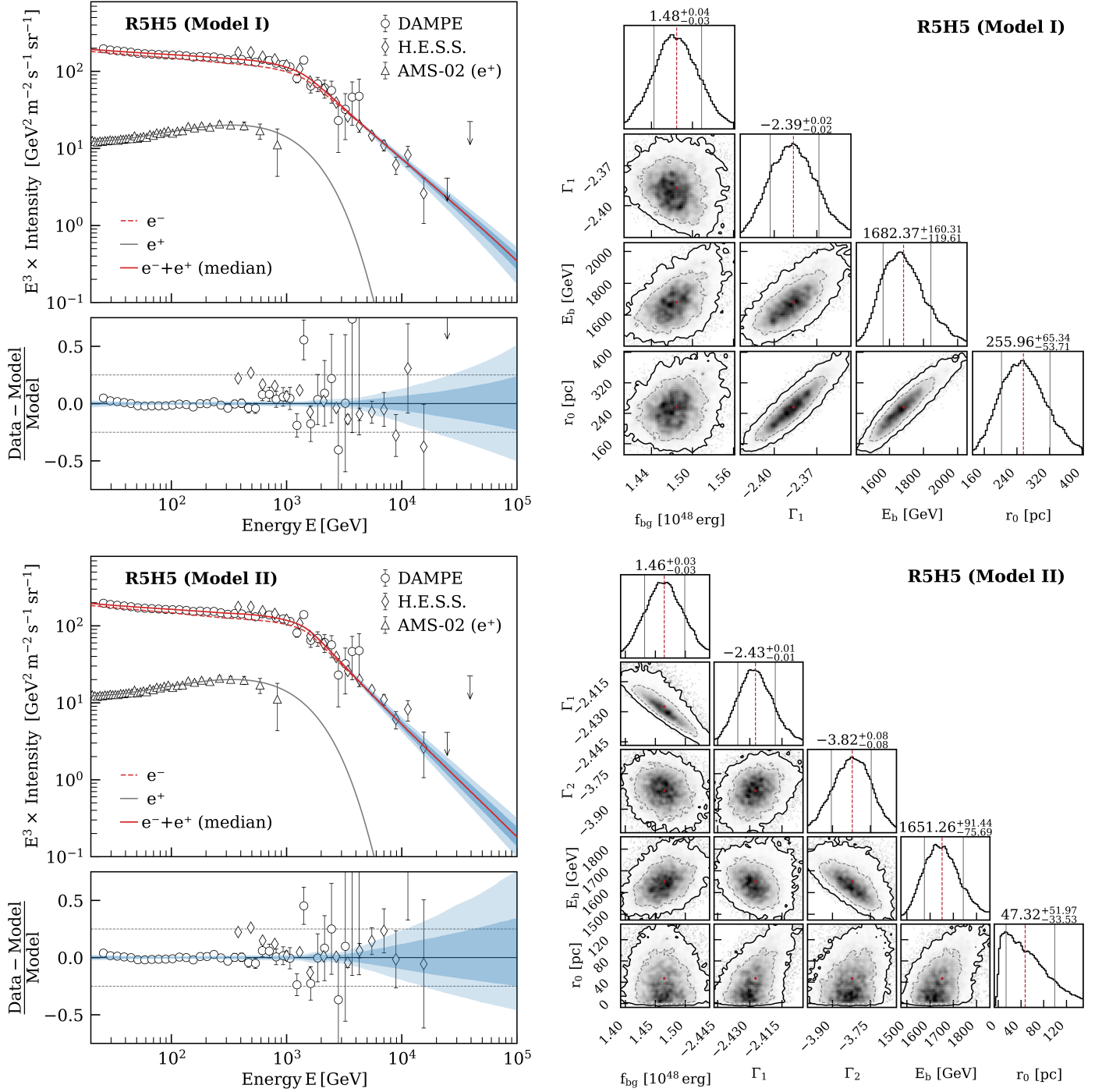


Fig. 4: Background fits and posterior distributions for the averaging region R5H5. The left column show the background electron component from the fits for R5H5 (Model I) and R5H5 (Model II), respectively (red dashed lines), the positron component given by the empirical parametrization of the AMS-02 collaboration [Equation (21)] (grey solid lines), and the resulting all-electron spectrum $e^- + e^+$ (red solid lines). Blue shaded bands show the 1σ and 2σ credible intervals of the all-electron spectrum. Data: DAMPE ($\circ$), H.E.S.S. ($\diamond$), and AMS-02 positrons ($\triangle$). Downward arrows denote upper limits from H.E.S.S. The lower panels in the left column show the residuals with respect to the median prediction of the total spectrum, where the dotted horizontal lines are guiding lines at ± 0.25 , and the blue bands indicate the corresponding 1σ and 2σ credible intervals. Fits to the all-electron $e^- + e^+$ spectrum in the range 20 GeV–5 TeV yield a reduced chi-squared of $\chi^2_\nu = 1.89$ ($\chi^2 = 77.30$ for 41 degrees of freedom) for Model I and $\chi^2_\nu = 1.11$ ($\chi^2 = 44.21$ for 40 degrees of freedom) for Model II. In both figures, the right column shows the corner plots of the marginalized posterior distributions of the model parameters: top panel for Model I, with (f_{bg} , Γ_1 , E_b , r_0), and bottom panel for Model II, with (f_{bg} , Γ_1 , Γ_2 , E_b , r_0). Diagonal panels show the one-dimensional marginalized posteriors, with red dashed lines indicating the medians and black dashed lines marking the 1σ credible interval. Off-diagonal panels show the two-dimensional marginalized posterior distributions. The solid and dashed contours enclose the 1σ and 2σ credible regions, respectively, and the red dot marks the posterior median values.

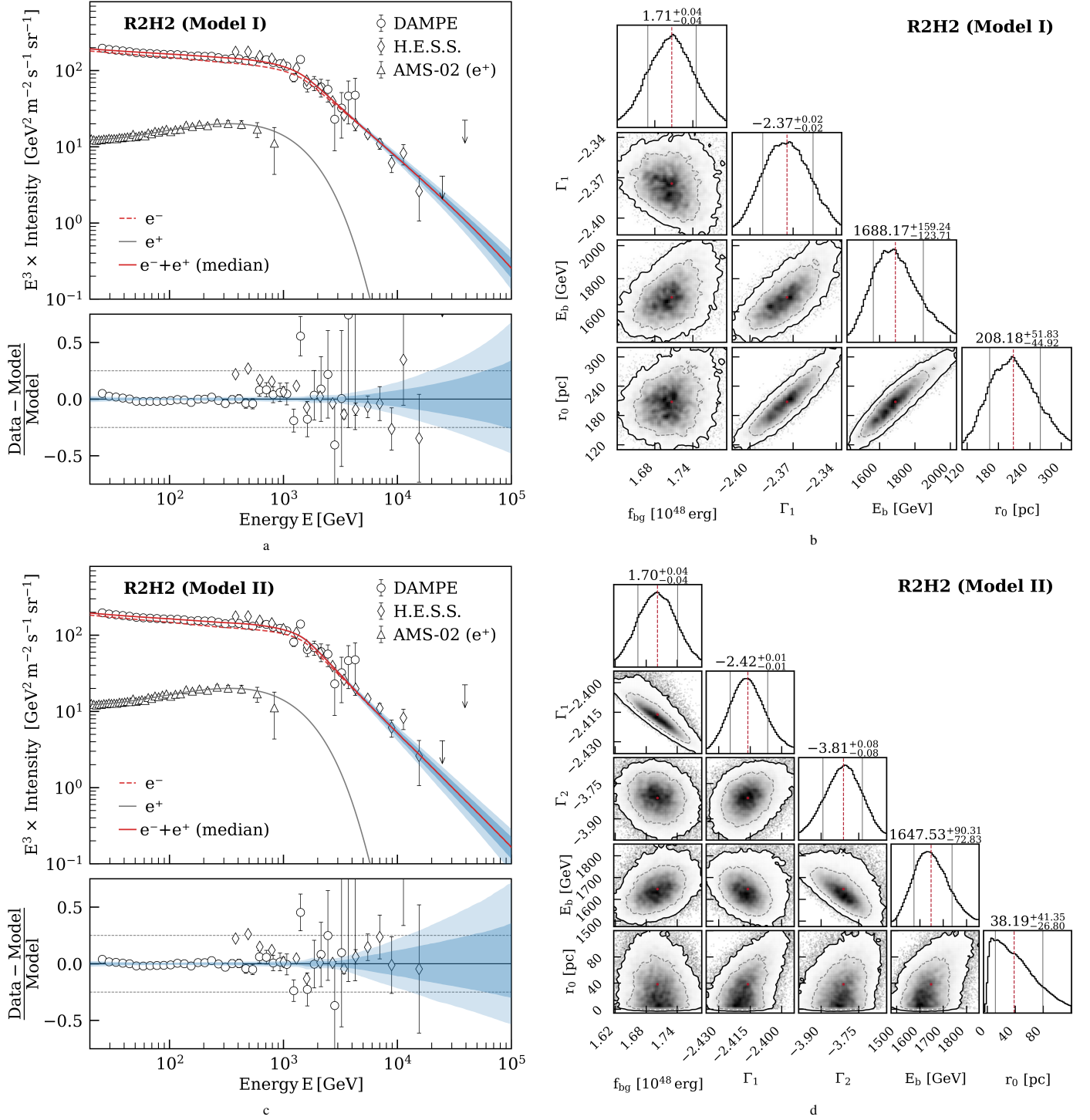


Fig. 5: Same as Fig. 4, but for R2H2 case. Fits to the all-electron $e^- + e^+$ spectrum in the range 20 GeV–5 TeV yield a reduced chi-squared of $\chi^2_\nu = 1.88$ ($\chi^2 = 77.1$ for 41 degrees of freedom) for Model I and $\chi^2_\nu = 1.11$ ($\chi^2 = 44.21$ for 40 degrees of freedom) for Model II. The right column shows the corresponding corner plots of the marginalized posterior distributions for Model I and Model II, respectively.

where the red dashed line denotes the median background electron spectrum obtained from 1000 samples⁵ drawn from the posterior distributions, the grey solid line the positron spectrum from the parametrization of the AMS-02 measurements, and the

red solid line represents the sum of the background electron and positron spectra. The shaded bands show the 1σ and 2σ credible intervals of the predicted flux. It may be noted that while the fit itself has been performed from 20 GeV to 5 TeV, the predicted spectra are plotted up to 100 TeV. The fit residuals are calculated with respect to the predicted background electron plus positron spectrum (red solid line), and the dotted horizontal lines in the residual plots at ± 0.25 level are shown only for reference. The

⁵ A posterior sample is a random draw from the posterior distribution of the model parameters, representing updated beliefs after observing the data, obtained from the MCMC.

right panels of Figures 4 and 5 show the corner plots of the fitted parameters. The diagonal plots display the one-dimensional marginalized posterior distributions (histograms), with the red dashed line indicating the median value and the black dashed lines marking the 1σ credible interval. The off-diagonal plots represent the two-dimensional marginalized distributions of posterior pairs with the contours indicating the 1σ and 2σ credible regions.

For the R5H5 case, the fit with Model I results in a reduced chi-squared value of $\chi^2_\nu = 1.89$ ($\chi^2 = 77.30$ with 41 degrees of freedom), while the fit with Model II yields $\chi^2_\nu = 1.11$ ($\chi^2 = 44.21$ with 40 degrees of freedom). For Model I, the posterior median values of the model parameters are $f_{bg} = 1.48^{+0.04}_{-0.03} (\times 10^{48})$ ergs for the electron injection efficiency, $\Gamma_1 = -2.39^{+0.02}_{-0.02}$ for the electron source spectral index below the break, $E_b = 1682^{+160}_{-120}$ GeV for the break energy, and $r_0 = 256^{+65}_{-54}$ pc for the minimum distance to the background sources. For Model II, the values of the electron injection efficiency and the break energy are similar to those obtained for Model I, with $f_{bg} = 1.46^{+0.03}_{-0.03} (\times 10^{48})$ erg and $E_b = 1651^{+91}_{-76}$ GeV. However, Model II gives a steeper source index below the break, $\Gamma_1 = -2.43^{+0.01}_{-0.01}$, along with an index above the break of $\Gamma_2 = -3.82^{+0.08}_{-0.08}$, compared to Model I. The spectrum thus steepens by $\Delta\Gamma \equiv \Gamma_2 - \Gamma_1 \approx -1.4$, which is larger than the $\Delta\Gamma = -1$ adopted in Model I. Moreover, Model II gives a smaller median r_0 value of $47.3^{+52.0}_{-33.5}$ pc. The corresponding best-fit values of the model parameters are $(f_{bg}, \Gamma_1, E_b, r_0) = (1.46 \times 10^{48} \text{ erg}, -2.39, 1625 \text{ GeV}, 232 \text{ pc})$ for Model I and $(f_{bg}, \Gamma_1, \Gamma_2, E_b, r_0) = (1.47 \times 10^{48} \text{ erg}, -2.43, -3.84, 1615 \text{ GeV}, 3.55 \text{ pc})$ for Model II. Except for the r_0 value in Model II, the best-fit and the median values of the parameters are similar as expected from their posterior distributions shown in Figures 4 and 5 (right panels) which are close to Gaussian. The posterior distribution of r_0 in Model II is strongly skewed towards its lower prior limit which, together with its best-fit value of 3.55 pc (close to zero), indicates that the data are consistent with $r_0 \rightarrow 0$. This can be understood, as Γ_2 is allowed to vary freely in Model II, the fit can reproduce the observed spectral steepening above the break by adjusting Γ_2 , without depending much on r_0 . In Model I, as Γ_2 is already fixed at $\Gamma_1 - 1$, the fit demands a stronger dependence on r_0 (as source spectrum has less freedom) in order to reproduce the observed spectral steepening. The corner plots in Figures 4 and 5 show strong positive correlation between r_0 and both E_b and Γ_1 for Model I. This indicates that an increase in r_0 must be compensated by a harder source spectral index and a larger break energy E_b . This behaviour is expected because increasing r_0 effectively suppresses the contribution from nearby high-energy sources. This suppression steepens the propagated spectrum at high energies and therefore has to be compensated by hardening the source spectrum and shifting the break to higher energies. In contrast, for Model II, where Γ_2 is treated as a free parameter, the observed TeV break can be reproduced largely through the intrinsic broken power-law source spectrum and the dependence on r_0 is much weaker. The anti-correlation between E_b and Γ_2 in the corner plot is characteristic of a broken power-law: increasing the break energy and injecting more high-energy electrons must be compensated by a steeper high-energy index to keep the predicted highest-energy flux consistent with the observed flux at the highest energies. Similarly, the anti-correlation between f_{bg} and Γ_1 reflects that an increase in the background normalisation, which increases the total low-energy flux, must be compensated by a steeper

source index to maintain the observed spectral slope below the break. [Here, add few lines on the behavior of the correlation plots for Model I and Model II. -done]

The R2H2 case yields $\chi^2_\nu = 1.88$ for Model I ($\chi^2 = 77.1$, 41 degrees of freedom) and $\chi^2_\nu = 1.11$ for Model II ($\chi^2 = 44.21$, 40 degrees of freedom), indicating a fit quality similar to that of the R5H5 case. The posterior median values for Model I and Model II agree within 1σ with those obtained in the R5H5 case, except for the electron injection efficiency (see Table 2). Furthermore, the best-fit values of the model parameters closely match the posterior median values except for r_0 as in the R5H5 case. In addition, the posterior correlation plots show similar behavior in both the R5H5 and R2H2 cases. The injection efficiency f_{bg} is larger by a factor of ~ 1.15 in R2H2, which is due to higher radiative losses of electrons resulting from the stronger average magnetic and radiation fields, thereby requiring a higher electron injection efficiency to reproduce the observed flux. Similar to the R5H5 case, R2H2 gives a larger r_0 value of 208^{+52}_{-45} pc for Model I compared to Model II, which only weakly constrains the r_0 value at $38.19^{+41.35}_{-26.80}$ pc (best-fit value = 1.88 pc), compatible with $r_0 \rightarrow 0$. Based on the r_0 values obtained for both the R5H5 and R2H2 cases, only Model I is largely consistent with the presence of nearby sources. Therefore, in the following, we focus exclusively on Model I for both cases to investigate the maximum possible contribution of nearby supernova remnants to the observed CR electron spectrum.

Table 3: Supernova remnants with distances within 300 pc considered in this work. References: (a) Egger & Aschenbach (1995), (b) Faherty et al. (2007), (c) Dodson et al. (2003), (d) Plucinsky et al. (1996).

SNR name	Distance (pc)	Age (yr)	References
Loop I	170	2.0×10^5	a
Geminga	250	3.4×10^5	b
Vela	287	1.1×10^4	c
Monogem	300	6.8×10^4	d

4.2. Maximum contribution of nearby supernova remnants

Among the R5H5 and R2H2 scenarios for the electron energy loss rates investigated in this work, we consider R5H5 to be more realistic because the halo size of $H = 5$ kpc is consistent with the halo boundary adopted in the calculation of the boron-to-carbon ratio (see Section 2.1). Therefore, based on the R5H5 case (Model I), which gives a median r_0 range of (202 – 321) pc, we choose $r_0 = 300$ pc as the distance to the nearest background cosmic-ray sources. All sources located within 300 pc are treated using a discrete space-time model as described in Section 2.5. From the literature, we identify four SNRs located within 300 pc radius from the Earth. They are listed in Table 3 together with their distances and ages.

For consistency, we consider the same source index Γ_1 and break energy E_b for the nearby sources as those obtained from the background fit. Since the best-fit values correspond to a single parameter set, whereas the median values are computed from 1000 posterior samples, and for Model I, as the best-fit and the median values are similar, we use the best-fit values of Γ_1 and E_b from the background fit for all subsequent calculations. We treat the electron injection efficiency f_{near} and the CR escape time parameter α of the nearby sources as free parameters, as described

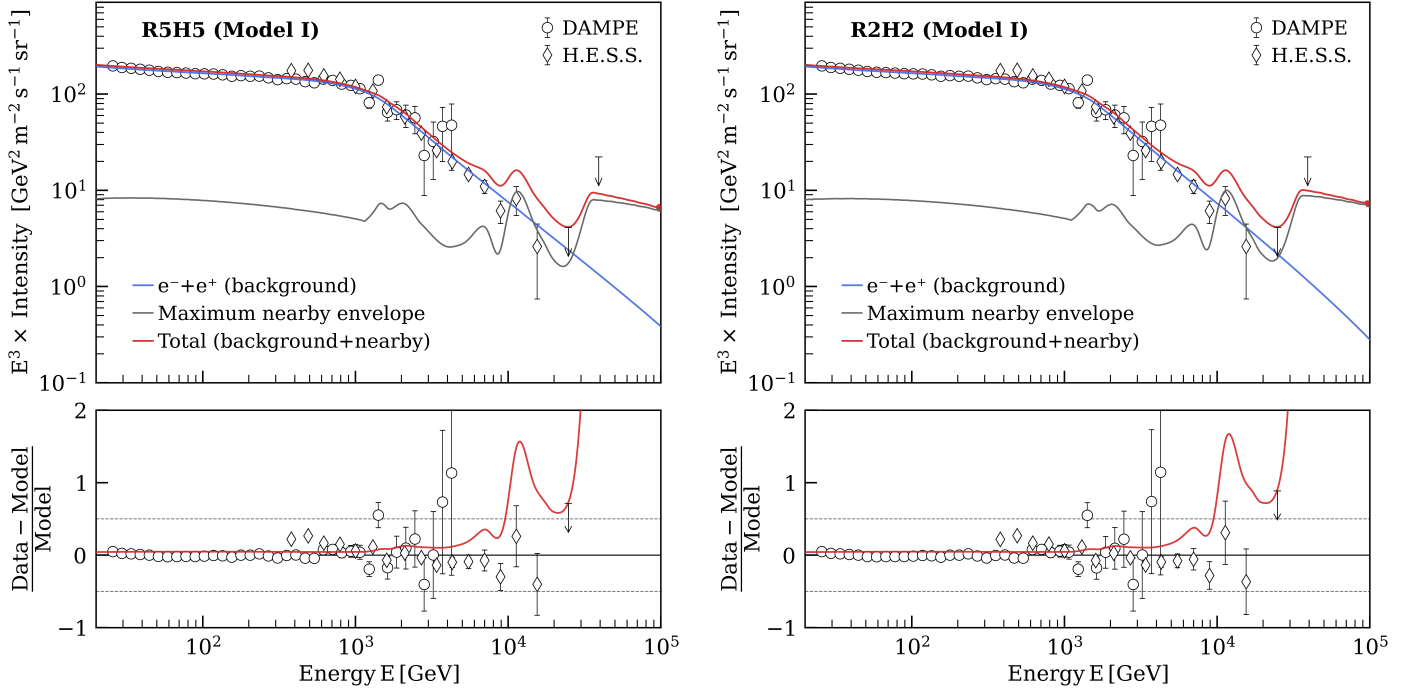


Fig. 6: Results for Model I (R5H5 in the left panels and R2H2 in the right panels). In the top panels, the blue curves show the background all-electron component ($e^- + e^+$) obtained from the MCMC best fit, while the grey curves denote the maximum nearby contribution envelope. The red curves represent the resulting maximum total electron envelope obtained as the sum of the background (blue) and maximal nearby (grey) components. Data: DAMPE ($\circ$) and H.E.S.S. ($\diamond$), with downward arrows indicating upper limits from H.E.S.S. The lower panels show the residuals with respect to the background electron flux, and dotted horizontal lines are guiding lines at ± 0.5 .

in Section 3.2, and consider all nearby SNRs to have the same source parameters.

For different α values in the range of 0.2 – 3.0, the maximum contribution of the nearby SNRs is obtained by adjusting f_{near} such that the sum of the nearby and background electron fluxes remains within the 3σ upper bound of the DAMPE and H.E.S.S. data at any energy. Figure 6 shows the results of the analysis for the R5H5 (left panel) and R2H2 (right panel) cases where the black line represents the maximum nearby flux, the blue line represents the background flux (best-fit electron flux plus positron flux), and the red line the sum of the nearby and background fluxes. The lower panels show the residual flux calculated with respect to the background. [I will look into the rest later]

The maximum nearby flux consistent with the 3σ bound for different values of the escape-time index α is shown in Figure 7, where each colored curve represents the maximum allowed nearby contribution and the color bar indicates the corresponding value of α . The solid black line traces the pointwise maximum of these curves at each energy point and is defined as the maximum nearby envelope adopted throughout our analysis. Because the total nearby flux corresponding to different values of α dominates at different energies, the maximum nearby envelope cannot be associated with a single physical realisation. Instead, it represents an α -independent upper bound, defined at each energy as the maximum over all possible nearby realisations.

[You need to discuss Figure 7 first (I don't even see any discussion on Figure 7 which is our most important figure). Then, discuss Figure 6. Actually, you should even put Figure 7 first.]

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Figure 6 shows how the maximum nearby envelope shapes the total $e^- + e^+$ spectrum. In each panel, the blue line shows the MCMC best-fit background, the grey line the nearby envelope, and the red line their sum. By construction, this red curve represents the highest total $e^- + e^+$ flux that any allowed nearby configuration can produce. All physically permitted nearby realisations must produce a total flux that lies at or below this curve. For both R5H5 and R2H2 for Model I, it is evident from Figure 6 that the flux below 1 TeV is dominated by the background electrons, while the contribution from nearby sources remains negligible. At higher energies the nearby envelope begins to approach the background electron flux.

4.3. Dominant nearby SNRs and their energy ranges

As shown in Figure 7, different values of α set the maximum nearby envelope at different energies. This naturally motivates an investigation of which value of the escape-time index α sets the nearby envelope at a given energy, which individual remnant dominates in that configuration, and what maximum injected energy budget is required. In this subsection, we address these issues by resolving the nearby envelope into source-specific contributions.

We construct source-resolved envelopes by following the same procedure used for the maximum total nearby contribution, but applied to each individual SNR. From the set of envelope-compatible realisations ($\alpha, f_{\text{near}}(\alpha)$), we resolve the total nearby contribution into Vela, Geminga, Monogem, and Loop I for each

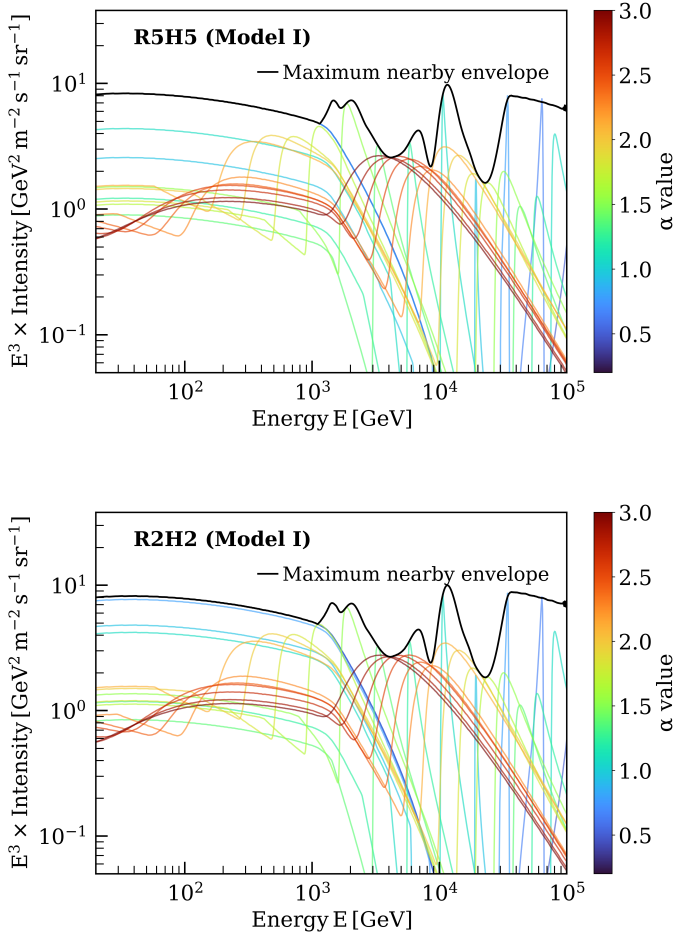


Fig. 7: Nearby-only electron spectra for different values of the escape-time index α , with the top panel showing R5H5 (Model I) and the bottom panel showing R2H2 (Model I). Each colored curve shows the propagated nearby contribution for a given α . At each energy, the maximum over all α values defines the black envelope curve, which provides an α -independent upper bound on the nearby contribution.

α . For a given energy E , the maximum allowed contribution of a given remnant is then defined as the largest flux it attains across all envelope-compatible values of α , yielding a source-resolved maximum contribution envelope for each SNR. These envelopes for both R5H5 (Model I) and R2H2 (Model I) are shown in Figure 8, where the solid coloured curves represent the maximum allowed flux from each remnant (Vela in red, Geminga in blue, Monogem in green, and Loop I in purple), and the shaded bands indicate the energy ranges over which the corresponding source can dominate the nearby contribution for at least one envelope-compatible value of the escape-time index. Here, dominance at a given energy means that the maximum-contribution envelope of that SNR lies above those of all other nearby SNRs at that energy. The structure of the source-resolved envelopes depends on the escape-time index α and on the ages and distances of the nearby SNRs. For the older remnants Loop I and Geminga ($t > 10^5$ yr), particle release is governed by the escape time $T_{\text{esc}}(E)$ as defined in Sec. 2.5. For such old remnants, the highest-energy electrons escape earliest and experience the strongest diffusive and radiative losses during propagation, so the surviving flux is dominated by lower-energy electrons, which therefore set

the observational constraints. In contrast, the contributions from the younger and intermediate-age remnants Vela and Monogem depend sensitively on α . For small α their contribution is confined to narrow features at the highest energies, whereas larger values of α shift and broaden these features toward lower energies. Selected examples illustrating how varying α affect the shape and composition of the nearby spectrum are shown in Appendix C. As a result, the nearby contribution below 1 TeV is primarily constrained by Loop I. At higher energies, the dominant constraints are instead set by Monogem and Vela, as their spectral features enter the energy range probed by DAMPE and H.E.S.S. (20 GeV–40 TeV).

At each energy E , we identify the escape index $\alpha(E)$ whose nearby contribution sets the envelope and determine the corresponding maximal injection fraction $f_{\text{near}}^{\text{max}}(E)$ for the dominant SNR. The resulting dominant escape index $\alpha(E)$ and maximal allowed injection fraction are shown in Figure 9, with the colour scale indicating its value. The bottom-left band represents Loop I together with the corresponding $f_{\text{near}}^{\text{max}}$. The diagonal band is associated with Monogem, while the upper scattered points correspond to Vela, which appears in distinct energy ranges for both R5H5 (Model I) and R2H2 (Model I). Below 1 TeV, the envelope is entirely controlled by Loop I, and the corresponding maximal injected energy obtained for this SNR is $f_{\text{near}}^{\text{max}} = 6.52(\times 10^{47})$ erg for R5H5 (Model I) and $f_{\text{near}}^{\text{max}} = 6.32(\times 10^{47})$ erg for R2H2 (Model I). Monogem can dominate over a broad intermediate energy range. For R5H5 (Model I), the envelope-compatible values of the escape index for Monogem span $\alpha \approx 0.54$ –1.69, with corresponding maximal injected energies $f_{\text{near}}^{\text{max}} \approx (0.62$ – $6.52)(\times 10^{47})$ erg, while for R2H2 (Model I) they span $\alpha \approx 0.54$ –1.71, with $f_{\text{near}}^{\text{max}} \approx (0.59$ – $6.32)(\times 10^{47})$ erg. Vela can dominate at the highest energies. For R5H5 (Model I), this occurs in two distinct intervals around $E \sim 10$ TeV and 20–30 TeV, corresponding to two clusters of dominant escape indices, one with $\alpha \approx 2.17$ –2.32 and $f_{\text{near}}^{\text{max}} \approx (0.70$ – $0.87)(\times 10^{47})$ erg, and a second with $\alpha \approx 1.60$ –2.01 and $f_{\text{near}}^{\text{max}} \approx (0.82$ – $1.22)(\times 10^{47})$ erg. For R2H2 (Model I), Vela dominates in similar energy intervals, with escape indices $\alpha \approx 2.14$ –2.34 and $f_{\text{near}}^{\text{max}} \approx (0.70$ – $1.00)(\times 10^{47})$ erg in the first cluster, and $\alpha \approx 1.60$ –2.03 with $f_{\text{near}}^{\text{max}} \approx (0.84$ – $1.26)(\times 10^{47})$ erg in the second cluster. In all envelope-compatible configurations, $\alpha(E)$ decreases systematically with increasing energy, reflecting the fact that smaller escape indices shift the contributions of younger SNRs toward higher energies. Above 1 TeV, the maximal nearby contribution is always determined by either Monogem or Vela, with allowed injected energy budgets spanning $f_{\text{near}}^{\text{max}} \approx 6.2(\times 10^{46})$ – $6.52(\times 10^{47})$ erg for R5H5 (Model I) and $f_{\text{near}}^{\text{max}} \approx 5.9(\times 10^{46})$ – $6.32(\times 10^{47})$ erg for R2H2 (Model I).

4.4. Effect on the dipole anisotropy

Nearby sources that contribute to the local CR electron spectrum are also expected to produce a dipole anisotropy. If a single nearby source dominates the flux at energy E , the anisotropy can be written (Thoudam 2007) as

$$\delta(E) = \frac{I_{\text{dom}}(E)}{I_{\text{tot}}(E)} \delta_{\text{dom}}(E), \quad (24)$$

where I_{dom} is the intensity from the dominant nearby source and I_{tot} is the total intensity. Under the diffusion approximation, the contribution from the dominant source is given by Mao & Shen (1972)

$$\delta_{\text{dom}}(E) = \frac{3D(E)}{c} \frac{|\nabla N_{\text{dom}}(E)|}{N_{\text{dom}}(E)}, \quad (25)$$

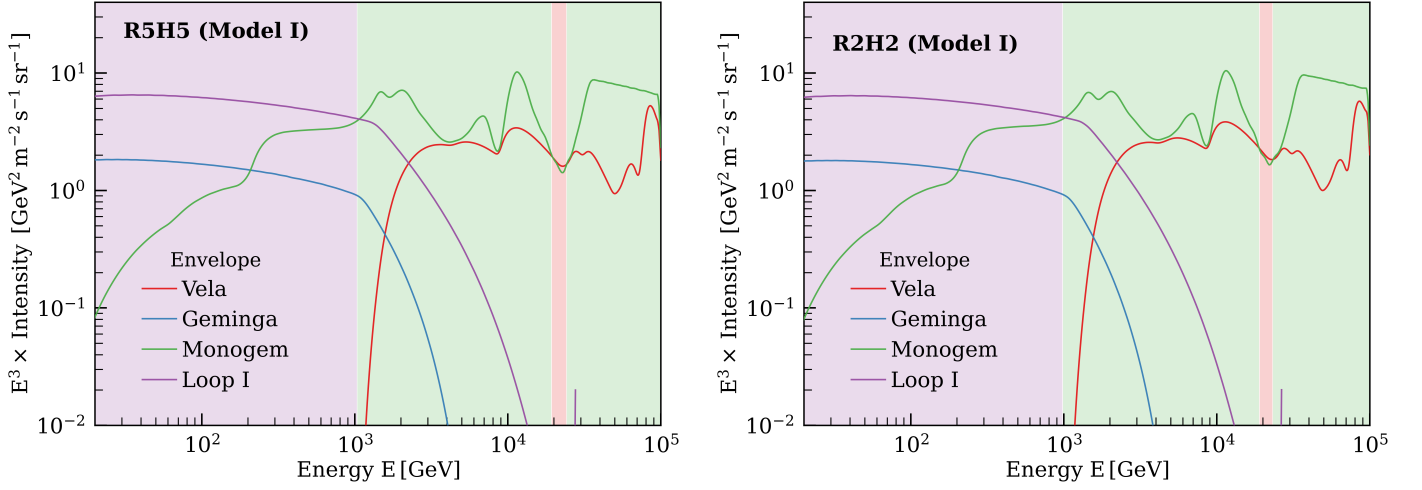


Fig. 8: Maximal nearby contributions of individual supernova remnants for Model I. The maximum contribution envelopes of Vela (red), Geminga (blue), Monogem (green), and Loop I (purple) are shown for R5H5 (left) and R2H2 (right). For each remnant, the envelope defines the maximal allowed flux across envelope-compatible realisations of the spectral index α . The shaded bands, shown in red for Vela, blue for Geminga, green for Monogem, and purple for Loop I, indicate the energies at which the corresponding remnant can act as the dominant nearby contributor for some envelope-compatible realisation of α . The corresponding dominant $\alpha(E)$ configurations are shown in Figure 9.

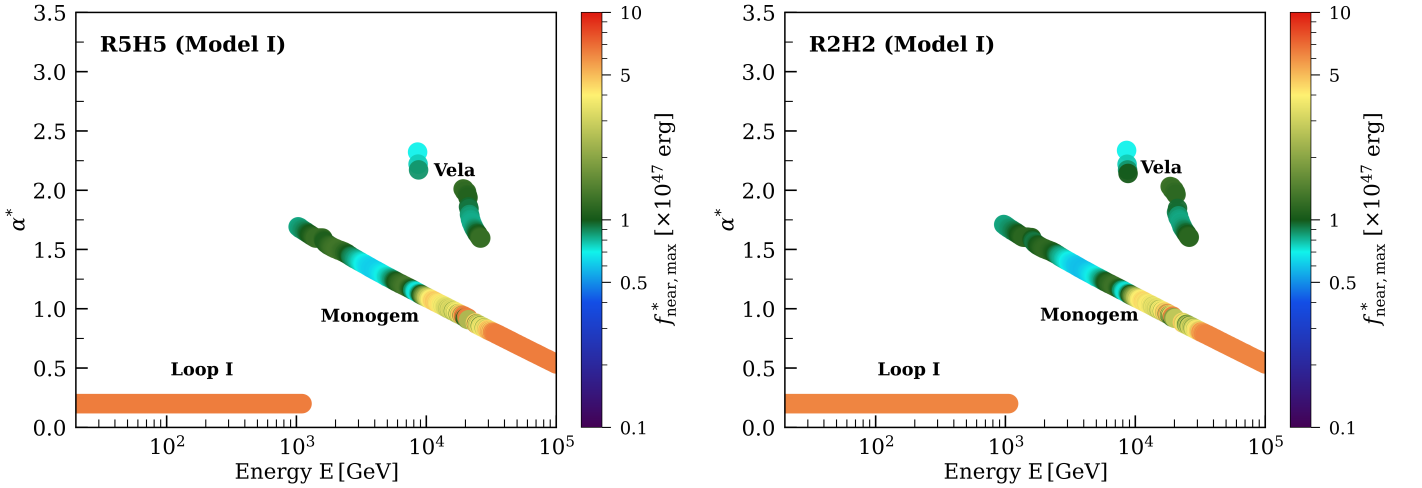


Fig. 9: Scatter plots of the dominant nearby source for Model I: R5H5 (left) and R2H2 (right). Each point shows the escape index $\alpha(E)$ and the corresponding maximal allowed injection fraction $f_{\text{near}}^{\text{max}}(E)$ of the source providing the dominant nearby contribution. The colour bar indicates the values of $f_{\text{near}}^{\text{max}}(E)$. The bottom-left band is associated with Loop I and the diagonal band with Monogem. The upper points correspond to Vela, which appears scattered in distinct energy ranges.

where N_{dom} is the propagated electron density from a source at distance r_{dom} and age t_{dom} , and $D(E)$ is the diffusion coefficient, taken to be the same as defined in Section 2.1.

In Section 3.2 we determined the maximal nearby contribution to the all-electron spectrum by scanning over the escape index α and the corresponding maximal injection fractions $f_{\text{near}}^{\text{max}}(\alpha)$. For each of these realisations we compute $\delta(E)$ from the above expressions for R5H5 (Model I) and R2H2 (Model I), and at each energy we retain the largest value over all realisations. This defines a maximal anisotropy envelope, i.e. the largest dipole amplitude compatible with the maximal nearby contribution to the spectrum. The resulting envelopes for R5H5 (Model I) and R2H2 (Model I) in are shown in Figure 10. Over the full energy range where measurements exist, the predicted anisotropy

lies well below the current 95% CL upper limits from *Fermi*-LAT (Abdollahi et al. 2017b) and CALET (Moz et al. 2025), independent of averaging region or background model. Thus the maximal nearby contribution inferred from the all-electron spectrum is fully consistent with present anisotropy constraints, and current limits do not further restrict $f_{\text{near}}^{\text{max}}$ or the associated electron energy budgets of nearby SNRs.

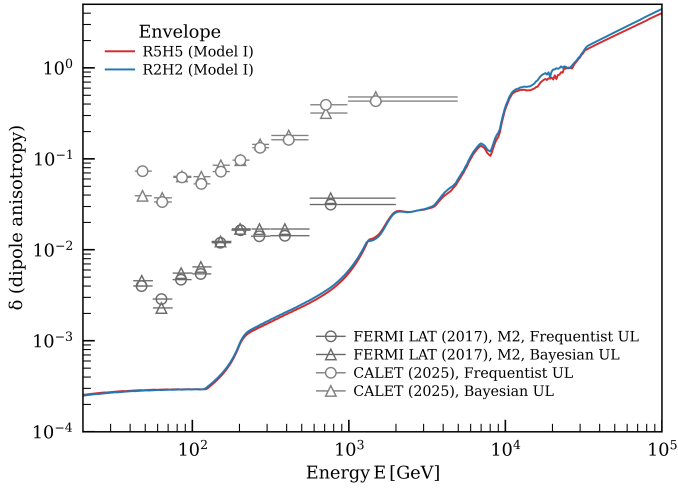


Fig. 10: Maximal dipole anisotropy envelope shown for R5H5 (Model I) in red and R2H2 (Model I) in blue. At each energy, the curves show the largest anisotropy obtained over all realisations of the nearby contribution. The resulting envelopes are compared with the 95% CL upper limits from the *Fermi*-LAT Method 2 (M2) dipole-anisotropy analysis (frequentist and Bayesian) (Abdollahi et al. 2017b) and from the CALET analysis (frequentist and Bayesian) (Motz et al. 2025).

5. Results and discussion

6. Conclusion

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Appendix A: Interpolation coefficients of the Klein–Nishina loss function

The coefficients c_i used in Equation (4) are listed in Table A.1. They correspond to the analytic interpolation of the Klein–Nishina loss function $Y(x)$ following the formulation of Fang et al. (2021).

Table A.1: Coefficients of the interpolating polynomial in Equation (4).

$c_0 = -3.996 \times 10^{-2}$	$c_1 = -9.100 \times 10^{-1}$
$c_2 = -1.197 \times 10^{-1}$	$c_3 = 3.305 \times 10^{-3}$
$c_4 = 1.044 \times 10^{-3}$	$c_5 = -7.013 \times 10^{-5}$
$c_6 = -9.618 \times 10^{-6}$	

Appendix B: Sedov–Taylor transition time

The Sedov–Taylor phase begins once the swept-up interstellar mass becomes comparable to the ejecta mass, marking the end of free expansion. The characteristic free-expansion velocity is

$$v_0 = \sqrt{\frac{2E_{\text{SN}}}{M_{\text{ej}}}} \approx 1.0 \times 10^4 \left(\frac{E_{\text{SN}}}{10^{51} \text{ erg}} \right)^{1/2} \left(\frac{M_{\text{ej}}}{M_{\odot}} \right)^{-1/2} \text{ km s}^{-1}, \quad (\text{B.1})$$

where E_{SN} is the explosion energy and M_{ej} the ejecta mass. During free expansion the shock radius evolves as $R \approx v_0 t$, so the swept-up ambient mass is $(4\pi/3)R^3 \mu m_p n$, where n is the ambient number density, μ the mean molecular weight, and m_p the proton mass. Equating the swept-up mass to the ejecta mass then yields the Sedov–Taylor transition time,

$$t_{\text{sed}} = \left(\frac{3}{4\pi \mu m_p n} \right)^{1/3} \frac{M_{\text{ej}}^{5/6}}{(2E_{\text{SN}})^{1/2}}. \quad (\text{B.2})$$

In scaled form,

$$t_{\text{sed}} \approx 0.186 \text{ kyr} \left(\frac{M_{\text{ej}}}{M_{\odot}} \right)^{5/6} \left(\frac{E_{\text{SN}}}{10^{51} \text{ erg}} \right)^{-1/2} \times \left(\frac{n}{1 \text{ cm}^{-3}} \right)^{-1/3} \left(\frac{\mu}{1.4} \right)^{-1/3}, \quad (\text{B.3})$$

which corresponds to transition times of a few 10^2 – 10^3 yr for typical Galactic supernova parameters. For a core-collapse explosion with $E_{\text{SN}} = 10^{51}$ erg, $n = 1 \text{ cm}^{-3}$, $\mu = 1.4$, and $M_{\text{ej}} = 8 M_{\odot}$, this gives $t_{\text{sed}} \approx 1.0$ kyr and $v_0 \approx 3.5 \times 10^3 \text{ km s}^{-1}$, which we adopt as representative values for the nearby SNR population.

Appendix C: Contribution from nearby sources for different escape indices α

To illustrate how the escape parameter α affects the nearby-source contribution, we show the α -dependence of the nearby flux in Model I for the R5H5 averaging region. The parameter α controls the energy dependence of the particle escape time from a supernova remnant (SNR), following $t_{\text{esc}} \propto E^{-1/\alpha}$, and therefore determines the release history of cosmic rays of different energies into the interstellar medium.

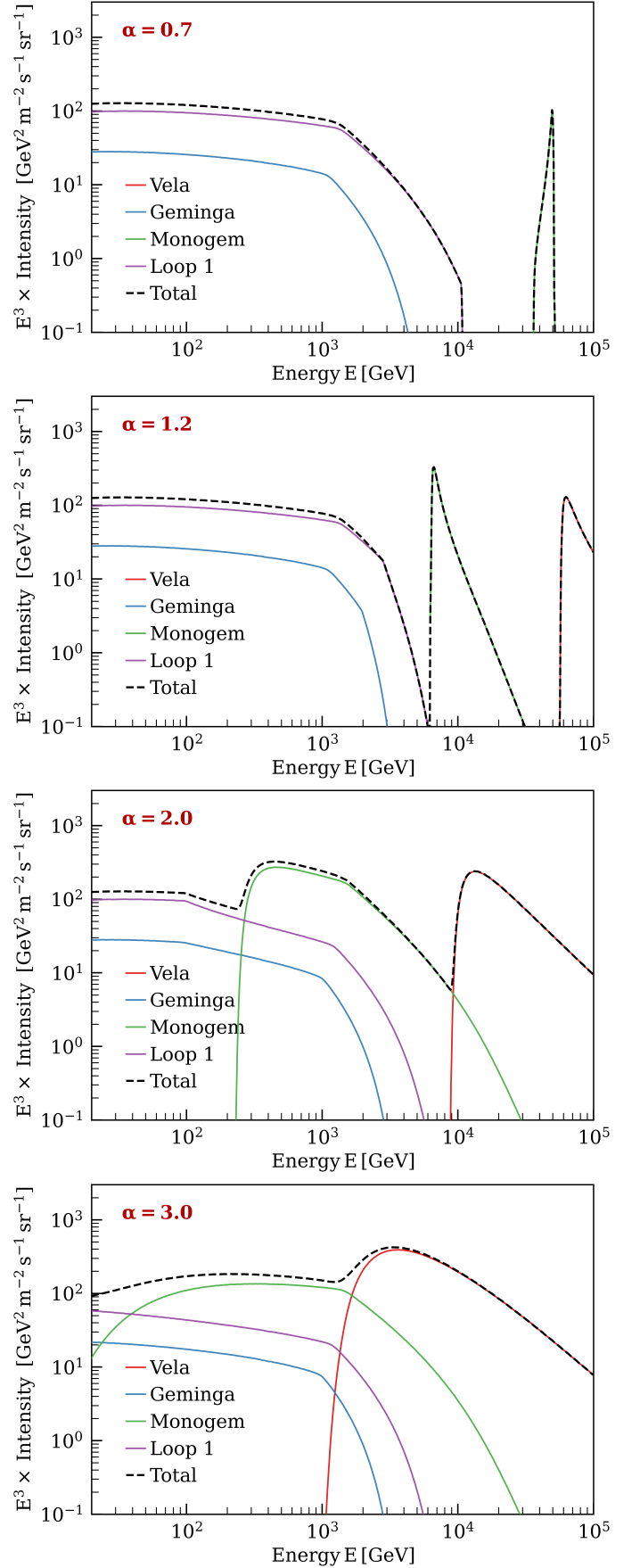


Fig. C.1: Relative contributions of nearby sources in Model I for the R5H5 region, shown for selected escape-index values α . Each panel shows the total nearby spectrum and individual components, illustrating how their relative contributions and combined spectrum vary with α .